TUTORIAL CHAPTER 3: MOTION IN TWO DIMENSIONS

3. A miniature quadcopter is located at $\mathbf{x_i=2.00\ m}$ and $\mathbf{y_i=4.50\ m}$ at $\mathbf{t=0}$ and moves with an average velocity having components $\mathbf{v_{av,x}=1.50\ m/s}$ and $\mathbf{v_{av,y}=-1.00\ m/s}$. At **t=2.00 s**, what are the

a. x-coordinate and

b. y-coordinate of the quadcopter's position?

6. A rabbit is moving in the positive x-direction at **2.00 m/s** when it spots a predator and accelerates to a velocity of **12.0 m/s** along the negative y-axis, all in **1.50 s**. Determine

a. the x-component and

b. the y-component of the rabbit's acceleration.

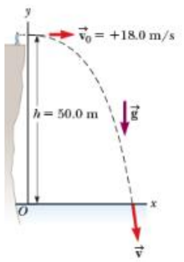

Figure P3.7

7. A student stands at the edge of a cliff and throws a stone horizontally over the edge with a speed of **18.0 m/s**. The cliff is **50.0 m** above a flat, horizontal beach as shown in **Figure P3.7**.

a. What are the coordinates of the initial position of the stone?

b. What are the components of the initial velocity?

c. Write the equations for the x- and y-components of the velocity of the stone with time.

d. Write the equations for the position of the stone with time, using the coordinates in **Figure P3.7**.

e. How long after being released does the stone strike the beach below the cliff?

f. With what speed and angle of impact does the stone land?

8. The fastest recorded pitch in Nippon Professional Baseball, thrown by Shohei Ohtani in 2016, was clocked at **101.9 mi/h**. If a pitch were thrown horizontally at this speed, how far would the ball fall vertically by the time it reached home plate, **60.5 ft** away?

9. The best leaper in the animal kingdom is the puma, which can jump to a height of **3.7 m** when leaving the ground at an angle of **45°**. With what speed must the animal leave the ground to reach that height?

10. A rock is thrown upward from the level ground in such a way that the maximum height of its flight is equal to its horizontal range **R**.

a. At what angle $\boldsymbol{\theta}$ is the rock thrown?

b. In terms of the original range **R**, what is the range $\mathbf{R_{max}}$ the rock can attain if it is launched at the same speed but at the optimal angle for maximum range?

c. Would your answer to part (a) be different if the rock is thrown with the same speed on a different planet? Explain.

12. Competitors in the Highland Games may participate in the Braemar Stone Throw, launching a **16.0−** to **22.0−**pound stone from a standing position. The event's current distance record of **17.4 m** was set in 1981 by Geoff Capes. Assuming the initial launch angle was **45.0°** and neglecting air resistance, determine

a. the initial speed of the projectile and

b. the total time the projectile was in flight.

c. Qualitatively, how would the answers change if the launch angle were greater than **45.0°**? Explain.

14. From the window of a building, a ball is tossed from a height $\mathbf{y_0}$ above the ground with an initial velocity of **8.00 m/s** and angle of **20.0°** below the horizontal. It strikes the ground **3.00 s** later.

a. If the base of the building is taken to be the origin of the coordinates, with upward the positive y −direction, what are the initial coordinates of the ball?

b. With the positive x -direction chosen to be out the window, find the x - and y - components of the initial velocity.

c. Find the equations for the x - and y -components of the position as functions of time.

d. How far horizontally from the base of the building does the ball strike the ground?

e. Find the height from which the ball was thrown.

f. How long does it take the ball to reach a point **10.0 m** below the level of launching?

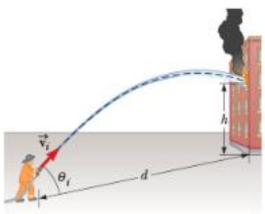

Figure P3.18

18. A fireman **d = 50.0 m** away from a burning building directs a stream of water from a ground-level fire hose at an angle of $\boldsymbol{\theta_i}$**= 30.0°** above the horizontal as shown in **Figure P3.18**. If the speed of the stream as it leaves the hose is $\mathbf{v_i = 40.0\ m/s}$, at what height will the stream of water strike the building?

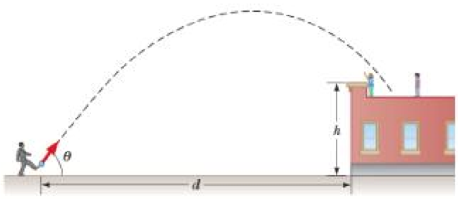

Figure P3.19

19. A playground is on the flat roof of a city school, $\mathbf{6.00\ m}$ above the street below (**Fig. P3.19**). The vertical wall of the building is $\mathbf{h = 7.00\ m}$ high, to form a **1**-m-high railing around the playground. A ball has fallen to the street below, and a passerby returns it by launching it at an angle of $\boldsymbol{\theta = 53.0}°$ above the horizontal at a point $\mathbf{d = 24.0\ m}$ from the base of the building wall. The ball takes **2.20 s** to reach a point vertically above the wall.

a. Find the speed at which the ball was launched.

b. Find the vertical distance by which the ball clears the wall.

c. Find the horizontal distance from the wall to the point on the roof where the ball lands.

21. Suppose a boat moves at **12.0 m/s** relative to the water. If the boat is in a river with the current directed east at **2.50 m/s**, what is the boat's speed relative to the ground when it is heading

a. east, with the current, and

b. west, against the current?

22. A car travels due east with a speed of **50.0 km/h**. Raindrops are falling at a constant speed vertically with respect to the Earth. The traces of the rain on the side windows of the car make an angle of **60.0°** with the vertical. Find the velocity of the rain with respect to

a. the car and

b. the Earth.

29. A river has a steady speed of **0.500 m/s**. A student swims upstream a distance of **1.00 km** and swims back to the starting point.

a. If the student can swim at a speed of **1.20 m/s** in still water, how long does the trip take?

b. How much time is required in still water for the same length swim?

c. Intuitively, why does the swim take longer when there is a current?

Answer:

3. a) 5.00 m, b) 2.50 m

6. a) -1.33 m/s^2, b) -8.00 m/s^2

7. a) x_0=0 and y_0= +50.0 m, b) u_x= +18.0 m/s and u_y= 0, c) v_x= 18.0 m/s and $v_y = -(9.80\ m/s^2)t$, d) $x = (18.0\ m/s)t$ and $y = 50.0\ m - (4.90\ m/s^2)t^2$, e) 3.19 s, f) 36.1 m/s at $60.1°$ below the horizontal.

8. -0.800 m

9. 12 m/s

10. a) $76.0°$, b) R_{max}= 2.13 R, c) The result is valid on every planet. Prove it from the formula.

12. a) 13.1 m/s, b) 1.89 s, c) Think in term of launch angle and launch speed.

14. a) (0,y_0), b) u_x =+7.52 m/s and u_y = -2.74 m/s, c) $x = (7.52\ m/s)t$ and $y = y_0 - (2.74\ m/s)t - (4.90\ m/s^2)t^2$, d) 22.6 m, e) $y_0 = 52.3\ m$, f) 1.18 s.

18. h=18.6 m

19. a) 18.1 m/s, b) 1.09 m, c) 2.8 m.

21. a) $|v_{BG}|$ =14.5 m/s, b) $|v_{BG}|$ =9.50 m/s.

22. a) $\vec{v}_{RC} = 57.7\ km/h\ at\ 60.0°\ west\ of\ vertical$ b) $\vec{v}_{RE} = 28.9\ km/h\ vertically\ downward$

29. a) 2.02×10^3 s, b) 1.67×10^3 s, c) Extra time required against the current.